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The Editors
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Dear Editors,

I submit the manuscript “Mechanistic Monte Carlo assessment of co-localized permeation enhancement and efflux modulation for oral peptide delivery” for consideration as a research article in the International Journal of Pharmaceutics.

Oral peptide products reach an absolute bioavailability of only about 1%, with high variability. The mechanistic literature on the best-characterized enhancer tablet shows that the effect is local and transient: enhancer, peptide and protection act only where and when they coincide. The manuscript asks, quantitatively, when a dosage-form architecture that deliberately co-localizes a reversible permeation enhancer, a P-glycoprotein modulator and a protease shield in a thin niche would raise bioavailability above 4% with reversible epithelial exposure, and when it would fail. It is a computational design-space study, and I have written it as one: it contains no experimental data, every probability is conditional on author-assigned, evidence-graded priors, and the reference arm is calibrated to a published baseline, so it is not presented as validation.

The study contributes a mechanistic model with every equation, unit and initial condition stated; paired Monte Carlo arms in which ablations and negative controls isolate co-localization, enhancer clearance, protection and efflux modulation; global variance-based sensitivity analysis with a decomposition of the output uncertainty; correlated priors; a benefit-risk analysis and falsification maps with predefined failure conditions; a structural-uncertainty analysis that replaces the lumped niche by stagnant-layer and spatial diffusion-reaction models; an observation-conditioned analysis; and comparison with published human bioavailability data that were not used for calibration.

The principal findings are conditional and, in part, unfavorable to the concept. With the reference calibrated to 1.2%, the Dual-Lock architecture reaches a median bioavailability of 5.0% at half the enhancer dose and meets the target with reversible exposure in 45% of draws against 15% for the reference; confinement accounts for most of the benefit, but the benefit disappears if the enhancer is not self-limiting. The model's advantage persists in every transport structure that can reproduce the published baseline, but it falls to 22% if the reference tablet performs like oral semaglutide; the upper tail of the predicted bioavailability lies beyond any published value; the model does not reproduce the observed optimum enhancer dose; and the matrix-only negative control reveals an untested

assumption. The paper turns these results into measurable failure boundaries (for example an effective niche thickness above 0.6 cm) for the first experiments.

Code, tests, all numerical results and the Supplementary Information are openly available at https://github.com/sandlerleon/dual-lock-oral-peptide-mc and are archived at Zenodo (software, doi:10.5281/zenodo.23864330; manuscript preprint, doi:10.5281/zenodo.23864327). Fixed seeds regenerate every number. The original model was re-run from a clean directory before any change; its headline results reproduced exactly, and the one discrepancy (tie handling in a rank correlation, up to 0.03 in 18 sensitivity coefficients) is reported in the repository.

A generative AI assistant (Claude, Anthropic) was used to help develop and test the simulation code, to run the analyses and to draft and revise the manuscript text and figures, as declared in the manuscript. I specified the research questions, reviewed and edited all content and take full responsibility for it. The manuscript is original, has not been published elsewhere and is not under consideration at any other journal. I am the sole author, I have no competing interests, and the work received no specific funding. An earlier, non-peer-reviewed description of the concept was prepared as a proposal for an industry open-innovation challenge; the commercial content of that proposal is not part of this manuscript.

Thank you for considering this submission.

Sincerely,

Leon Sandler
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